

Coefficient Plot Reproduction

Chetty et al. (2014) Table VIII: Classical vs Permutation Inference

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This file contains only the data preparation, statistical tests, and plotting code needed to reproduce the coefficient plot from the two supplied CSV files.

```
# If needed, install the CRAN packages with:
# install.packages(c("ggplot2", "lmtest", "remotes", "sandwich"))
# remotes::install_github(
#   "hyu-ub/perk",
#   subdir = "perk",
#   ref = "067e48e01b8e761f2106aab7d98d470d831805ec"
# )
required_packages <- c("ggplot2", "lmtest", "perk", "sandwich")
missing_packages <- required_packages[
  !vapply(required_packages, requireNamespace, logical(1), quietly = TRUE)
]

if (length(missing_packages) > 0L) {
  stop(
    "Install the missing R packages before rendering: ",
    paste(missing_packages, collapse = ", ")
  )
}

find_data_file <- function(filename) {
  candidates <- unique(c(
    file.path(document_dir, filename),
    file.path(document_dir, "..", "data", filename),
    file.path(getwd(), filename),
    file.path(getwd(), "data", filename)
  ))
  matches <- candidates[file.exists(candidates)]

  if (length(matches) == 0L) {
    stop("Could not find ", filename, ".")
  }

  normalizePath(matches[[1]])
}

table5 <- read.csv(
  find_data_file("Online_Data_Table_5.csv"),
  skip = 49,
  stringsAsFactors = FALSE
)
table5 <- table5[
  !grepl("^\\((\\d+\\))$", table5[[1]]) & table5[[1]] != "",
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    ,
    drop = FALSE
]

table8 <- read.csv(
  find_data_file("Online_Data_Table_8.csv"),
  skip = 6,
  stringsAsFactors = FALSE
)
table8_cz <- suppressWarnings(as.numeric(table8[[1]]))
table8 <- table8[!is.na(table8_cz) & table8_cz > 0, , drop = FALSE]

names(table5) <- gsub("^X\\.", "", names(table5))
names(table8) <- gsub("^X\\.", "", names(table8))
names(table5)[c(1, 4, 6)] <- c("CZ", "n_children", "RM")
names(table8)[1] <- "CZ"

table5$CZ <- suppressWarnings(as.numeric(table5$CZ))
table8$CZ <- suppressWarnings(as.numeric(table8$CZ))

table8_match <- match(table5$CZ, table8$CZ)
matched_rows <- !is.na(table8_match)
mobility_columns <- c("CZ", "RM", "n_children")
characteristic_columns <- setdiff(names(table8), "CZ")

mobility_data <- table5[matched_rows, mobility_columns, drop = FALSE]
characteristic_data <- table8[
  table8_match[matched_rows],
  characteristic_columns,
  drop = FALSE
]
analysis_data <- cbind(mobility_data, characteristic_data)

numeric_columns <- c(
  "RM",
  "n_children",
  names(table8)[4:ncol(table8)]
)
analysis_data[numeric_columns] <- lapply(
  analysis_data[numeric_columns],
  function(value) suppressWarnings(as.numeric(value))
)
analysis_data <- analysis_data[
  analysis_data$n_children >= 250 & !is.na(analysis_data$RM),
  ,
  drop = FALSE
]
rownames(analysis_data) <- NULL

```

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permutation_regression_test <- function(
  x,
  y,
  permutations = 10000L,
  seed = NULL,
  batch_size = 250L) {
  if (!is.null(seed)) {
    set.seed(seed)
  }

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}

design <- cbind(1, as.numeric(x))
y <- as.numeric(y)
n <- nrow(design)

if (n != length(y)) {
  stop("x and y must have the same number of observations.")
}

# The design matrix is fixed across permutations, so these inverses are
# computed once rather than inside every permutation.
xtx_inverse <- solve(crossprod(design))
sigma_inverse <- n * xtx_inverse

compute_statistics <- function(response) {
  response <- as.matrix(response)
  beta <- xtx_inverse %*% crossprod(design, response)
  residual <- response - design %*% beta
  transformed_beta <- sigma_inverse %*% beta
  score <- design %*% transformed_beta
  colSums(score^2 * residual^2)
}

observed_statistic <- compute_statistics(y)[[1]]
exceedances <- 0L
remaining <- as.integer(permutations)

while (remaining > 0L) {
  current_batch <- min(batch_size, remaining)
  permuted_y <- replicate(
    current_batch,
    sample(y, replace = FALSE)
  )
  exceedances <- exceedances + sum(
    compute_statistics(permuted_y) >= observed_statistic
  )
  remaining <- remaining - current_batch
}

(exceedances + 1) / (permutations + 1)
}

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covariate_labels <- c(
  "Frac..Black" = "Fraction Black Residents",
  "Racial.Segregation" = "Racial Segregation",
  "Income.Segregation" = "Income Segregation",
  "Segregation.of.Poverty...p25." = "Segregation of Poverty (<p25)",
  "Segregation.of.Affluence...p75." = "Segregation of Affluence (>p75)",
  "Frac..with.Commute...15.Mins" = "Share with Commute < 15 Mins",
  "Household.Income.per.capita" = "Household Income per Capita",
  "Gini" = "Gini Coefficient",
  "Top.1..Income.Share" = "Top 1% Income Share",
  "Gini.Bottom.99." = "Gini Bottom 99%",
  "Frac..Between.p25.and.p75" = "Fraction Middle Class",
  "School.Expenditure.per.Student" = "School Expenditure per Student",
  "Teacher.Student.Ratio" = "Teacher-Student Ratio",

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"Test.Score.Percentile..Income.Adjusted" =
  "Test Score Percentile (Inc. Adj.)",
"High.School.Dropout.Rate..Income.Adjusted." =
  "HS Dropout Rate (Inc. Adj.)",
"Social.Capital.Index" = "Social Capital Index",
"Fraction.Religious" = "Fraction Religious",
"Violent.Crime.Rate" = "Violent Crime Rate",
"Fraction.of.Children.with.Single.Mothers" = "Fraction Single Mothers",
"Fraction.of.Adults.Divorced" = "Fraction Divorced",
"Fraction.of.Adults.Married" = "Fraction Married",
"Local.Tax.Rate" = "Local Tax Rate",
"Local.Government.Expenditures.Per.Capita" =
  "Local Gov Expenditures",
"State.EITC.Exposure" = "State EITC Exposure",
"State.Income.Tax.Progressivity" = "State Income Tax Progressivity",
"Number.of.Colleges.per.Capita" = "Colleges per Capita",
"College.Tuition" = "Mean College Tuition",
"College.Graduation.Rate..Income.Adjusted." =
  "College Grad Rate (Inc. Adj.)",
"Labor.Force.Participation.Rate" = "Labor Force Participation",
"Manufacturing.Employment.Share" = "Manufacturing Employment Share",
"Growth.in.Chinese.Imports.1990.2000" = "Growth in Chinese Imports",
"Teenage.Labor.Force.Participation.Rate" =
  "Teen Labor Force Participation",
"Migration.Inflow.Rate" = "Migration Inflow Rate",
"Migration.Outflow.Rate" = "Migration Outflow Rate",
"Frac..Foreign.Born" = "Fraction Foreign Born"
)

missing_covariates <- setdiff(names(covariate_labels), names(analysis_data))
if (length(missing_covariates) > 0L) {
  stop("Missing covariates: ", paste(missing_covariates, collapse = ", "))
}

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estimate_covariate <- function(index) {
  covariate <- names(covariate_labels)[[index]]
  model_data <- analysis_data[c("RM", covariate)]
  model_data <- model_data[complete.cases(model_data), , drop = FALSE]

  y <- as.numeric(scale(model_data$RM))
  x <- as.numeric(scale(model_data[[covariate]]))
  model <- lm(y ~ x)
  coefficient_table <- coef(summary(model))

  set.seed(42 + index)
  perk_result <- perk::perk_test(
    x,
    y,
    B = 1000,
    method = "pearson",
    alternative = "two.sided"
  )

  p_permutation_regression <- permutation_regression_test(
    x = x - mean(x),
    y = y - mean(y),
    permutations = 10000,

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    seed = 42 + index
  )

  data.frame(
    covariate = covariate,
    covariate_label = unname(covariate_labels[[covariate]]),
    n = nrow(model_data),
    beta = coefficient_table[2, "Estimate"],
    se = coefficient_table[2, "Std. Error"],
    p_classical = coefficient_table[2, "Pr(>|t|)"],
    p_correlation = cor.test(x, y, method = "pearson")$p.value,
    p_hc0 = lmtest::waldtest(
      model,
      vcov = sandwich::vcovHC(model, type = "HC0")
    )$`Pr(>F)`[2],
    p_hc3 = lmtest::waldtest(
      model,
      vcov = sandwich::vcovHC(model, type = "HC3")
    )$`Pr(>F)`[2],
    p_perk = perk_result$p.value,
    p_permutation_regression = p_permutation_regression
  )
}

results <- do.call(
  rbind,
  lapply(seq_along(covariate_labels), estimate_covariate)
)

p_value_columns <- c(
  "p_classical",
  "p_correlation",
  "p_hc0",
  "p_hc3",
  "p_perk",
  "p_permutation_regression"
)

significant_method_count <- rowSums(results[p_value_columns] < 0.05)
results$significance_agreement <- ifelse(
  significant_method_count == length(p_value_columns),
  "All significant",
  ifelse(
    significant_method_count == 0L,
    "None significant",
    "Disagreement"
  )
)

plot_data <- results[order(results$beta), , drop = FALSE]
plot_data$covariate_label <- factor(
  plot_data$covariate_label,
  levels = plot_data$covariate_label
)

coefficient_plot <- ggplot2::ggplot(
  plot_data,
  ggplot2::aes(x = beta, y = covariate_label)

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) +
ggplot2::geom_vline(
  xintercept = 0,
  linetype = "dashed",
  color = "gray50"
) +
ggplot2::geom_errorbar(
  ggplot2::aes(
    xmin = beta - 1.96 * se,
    xmax = beta + 1.96 * se,
    color = significance_agreement
  ),
  orientation = "y",
  width = 0.3,
  alpha = 0.7
) +
ggplot2::geom_point(
  ggplot2::aes(color = significance_agreement),
  size = 3
) +
ggplot2::scale_color_manual(values = c(
  "All significant" = "#2E8B57",
  "None significant" = "gray60",
  "Disagreement" = "#DC143C"
)) +
ggplot2::labs(
  title =
    "Predictors of Relative Mobility: Classical vs Permutation Inference",
  subtitle = paste(
    "Replication of Chetty et al. (2014) Table VIII with",
    "robustness check (6 methods)"
  ),
  x = "Standardized Coefficient (95% CI)",
  y = "",
  color = "Significance Agreement"
) +
ggplot2::theme_minimal() +
ggplot2::theme(
  legend.position = "bottom",
  axis.text.y = ggplot2::element_text(size = 9),
  plot.title = ggplot2::element_text(face = "bold"),
  plot.subtitle = ggplot2::element_text(color = "gray40")
)

plot_file <- file.path(document_dir, "coefficient_plot.png")
ggplot2::ggsave(
  plot_file,
  coefficient_plot,
  width = 10,
  height = 9,
  dpi = 300
)

coefficient_plot

```

Predictors of Relative Mobility: Classical vs Permutation Inference

Replication of Chetty et al. (2014) Table VIII with robustness check (6 methods)

